

ABHI TYAGI

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Professional Summary

MSc Business Analytics graduate (First Class Honours) with 2+ years of hands-on experience in Python, SQL, and analytics on a 500,000+ user US healthcare platform. Building end-to-end ML projects covering NLP sentiment classification and hybrid recommendation systems, deployed on AWS using Docker, MLflow, DVC, and GitHub Actions. Focused on building production-grade ML systems with reproducible pipelines and continuous validation.

Technical Skills

Languages: Python, SQL

Machine Learning: Supervised & Unsupervised Learning, Deep Learning, NLP, Recommendation Systems, Time-Series Forecasting, Explainable AI (SHAP)

Frameworks & Libraries: Scikit-learn, TensorFlow, LightGBM, Pandas, NumPy

MLOps: MLflow, DVC, Docker, GitHub Actions, Pytest, Model Registry, CI/CD

Cloud (AWS): S3, ECR, ECS/Fargate, EC2, IAM, CloudWatch

Analytics & Visualization: Statistical Analysis, EDA, Feature Engineering, Power BI, DAX, Matplotlib, Plotly, Streamlit, Flask

Databases: MySQL

Tools: Git, GitHub, Jupyter, REST APIs

Projects

YouTube Comment Sentiment Analysis

Python, LightGBM, NLTK, TF-IDF, Flask, MLflow, DVC, Docker, GitHub Actions, AWS (ECR, ECS/Fargate, S3, CloudWatch)

- Built a three-class sentiment classification system on YouTube comments using TF-IDF vectorisation and LightGBM, exposed via a Flask REST API and integrated into a Chrome extension for real-time inference on live videos.
- Implemented reproducible ML pipeline using DVC and MLflow model registry with candidate to champion alias promotion, gated by GitHub Actions CI running automated model loading, signature, and performance validation tests.
- Containerised the inference service with Docker, published to Amazon ECR, and deployed on Amazon ECS/Fargate with CloudWatch logging and IAM task roles for secure runtime AWS access.

Experience

Data Scientist

May 2025 – Present

Freelance

Remote

- Built a hybrid music recommendation engine combining content-based similarity and collaborative filtering with automatic content-based fallback, deployed as a Dockerised Streamlit application on Amazon EC2 with IAM instance profiles pulling private images from Amazon ECR.
- Designed a DVC-tracked reproducible data pipeline separating cleaning, collaborative filtering, and hybrid transformation stages, with artifacts stored on Amazon S3 and GitHub Actions CI validating dependencies and automated tests.
- Built ML models for stock price forecasting on 73,608 Reddit articles integrated with Yahoo Finance time-series data, applying Gradient Boosting, LSTM, CNN, and CNN-LSTM to achieve 56.1% accuracy and 57.1% ROC-AUC on S&P 500.
- Applied SHAP explainability to interpret model drivers and communicate predictions clearly to both technical and non-technical stakeholders.

Data Analyst

April 2022 – March 2024

Wipro Limited — Client: Alight Solutions

Noida, India

- Performed SQL and Python-based data extraction, validation, and analysis across a 500,000+ user healthcare platform, identifying trends, anomalies, and data quality issues to support operational and strategic decision-making.
- Built automated reporting workflows and KPI dashboards tracking pass rates, completion rates, and defect trends across quarterly release cycles, reducing manual effort by approximately 100 hours per quarter.
- Delivered analytical insights across 12-13 concurrent projects per release cycle, coordinating with cross-functional teams and translating requirements into structured outputs.
- Presented weekly analytical findings and data-driven recommendations to US-based senior stakeholders, translating complex outputs into clear business narratives.

Certifications

Oracle Certified Generative AI Professional • Oracle Certified OCI AI Foundations Associate • Data Science Mentorship Program (DSMP 2.0) — CampusX

Education

Dublin Business School

Dublin, Ireland

MSc in Business Analytics, First Class Honours

April 2024 – April 2025

Meerut Institute of Engineering and Technology (MIET)

India

B.Tech in Mechanical Engineering, First Division with Distinction

August 2018 – July 2022